A Project Report

on

DETECTION AND MITIGATION OF HALLUCINATIONS IN LARGE LANGUAGE MODELS

carried out as part of the ***Project Based Learning-2 (DSE2270)***

Submitted

by

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CERTIFICATE

This is to certify that the project-based learning, project titled **Detection and mitigation of hallucinations in large language models** is a record of the bonafide work done by **Aditi Singh** (2430010559) and **Dev Vashishtha** (2430010560), submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering (Data Science) of Manipal University Jaipur, Jaipur during the academic year 2025-26.

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ABSTRACT

Large Language Models (LLMs) have improved the field of natural language processing by allowing computers to respond to queries in a human-like fashion and helping in tasks like question-answering and information retrieval. The application of LLMs is on the rise in critical areas like healthcare, where AI can help medical professionals in quickly accessing information. However, LLMs tend to have hallucinations, which are answers that seem very plausible but are actually not true. In a healthcare setting, these answers that are not true can cause problems in decision-making.

The proposed project aims to detect hallucinations in Large Language Models for healthcare-related queries. The current state of the art tends to involve complex multi-model systems or computationally intensive verification processes. To overcome this problem, the proposed system will check the answers provided by LLMs for their correctness based on trusted medical knowledge sources and semantic similarity analysis. The system is developed using Python and tested with question-answer datasets for medical queries to improve the accuracy of AI-assisted healthcare systems.

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PROJECT BASED LEARING REPORT GUIDELINES

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1. Introduction

1.1 Background

Large Language Models (LLMs) like GPT, PaLM, and LLaMA led to major improvements in the field of natural language processing as they allow machines to write text as a human and respond to context. The models are very popular in many fields such as healthcare, education, customer support and research owing to their capability to handle vast volumes of data and give rapid responses.

In spite of their abilities, LLMs have a severe weakness in the form of hallucination, wherein model results are coherent and convincing but incorrect or unsupported by reality. Such imagined answers are usually hard to identify due to the fact that they closely resemble accurate data and the manner in which they are given is very confident.

The primary cause of hallucinations is that LLMs are trained to make the most probable word sequence prediction, as opposed to ensuring the accuracy of the information. Consequently, the model is reluctant to admit uncertainty when faced with uncertainty or when they do not have adequate knowledge, producing plausible answers. Also, the existence of noisy, biased, or outdated information in training data can also add to the wrong outputs.

The hallucinations can be presented in various forms, including errors of the facts, and fabricated information or contradictions within the response created. It is of particular concern in such a high-risk area as healthcare, where misinformation can have severe repercussions.

1.2 Purpose of the Document

The purpose of this document is to present a comprehensive study on the detection and mitigation of hallucinations in Large Language Models**.** This report aims to analyze the causes and types of hallucinations, review existing research solutions, and propose an effective approach to address this issue.

Additionally, the document focuses on designing and developing a prototype system that can identify hallucinated outputs and reduce their occurrence using suitable techniques. The study also evaluates the effectiveness of different detection and mitigation methods, particularly in the context of healthcare-related applications where accuracy and reliability are crucial.

Overall, this report serves as a structured guide to understanding the hallucination problem and exploring practical solutions to improve the trustworthiness of LLM-based systems.

1.3 Scope of the Project

This project will entail the research, design, and construction of a system to identify and counteract the hallucinations in the responses produced by LLM. The idea of the project is mainly to concentrate on healthcare related inquiries where the role of the misinformation is more important.
The system will summarize model results, detect possible hallucinations with methods that consist of consistency tests, detecting contradictions, and estimating confidence, and implement appropriate mitigation strategies such as retrieval-based verification and better prompting strategies. However, the project is limited to application only on the prototype level, and does not involve training gigantic models in itself. The latter is also the text-based LLA that does not elaborate on multimodal models too much. The findings are considered depending on the chosen datasets and they might not always be applicable in all real-world situations.

2. Literature Review

Title	Author(s)	Methods / Algorithms	Dataset	Key Findings	Limitations
The Dawn After the Dark: An Empirical Study on Factuality Hallucination in LLMs	J. Li et al.	HaluEval 2.0 benchmark, empirical detection & mitigation analysis across training stages	HaluEval 2.0	Hallucination arises from multiple stages (pretraining, RLHF); combined mitigation improves factuality	High computational complexity and large-scale evaluation required
Do Language Models Know When They're Hallucinating References?	A. Agrawal et al.	Consistency-based internal questioning and reference validation	Reference datasets	Models can identify hallucinated references via inconsistency in author/content details	Limited to citation/reference-based hallucination cases
Measuring Reliability of LLMs through Semantic Consistency	H. Raj et al.	Semantic consistency metrics across paraphrased prompts	TruthfulQA	Inconsistent responses across similar prompts strongly indicate hallucination	Requires multiple prompt evaluations, increasing cost
Self-Contradictory Hallucinations of LLMs	N. Mündler et al.	Self-contradiction detection and iterative refinement framework	Open-domain QA datasets	~17% generated outputs contain contradictions; removing contradictions improves reliability	Does not incorporate external factual verification
Trapping LLM Hallucinations Using Tagged Context Prompts	P. Feldman et al.	Context tagging and structured prompt engineering	Synthetic prompt datasets	Providing structured context significantly reduces hallucination (up to ~98%)	Highly dependent on prompt quality and design
Cognitive Mirage: A Review of Hallucinations in LLMs	H. Ye et al.	Taxonomy-based classification and theoretical analysis	Multiple datasets/tasks	Provides detailed taxonomy of hallucination types and causes	Primarily theoretical, lacks implementation
Survey of Hallucination Mitigation	S. Tonmoy et al.	RAG, CoVe, retrieval-based and hybrid	Various datasets	Identifies 30+ mitigation techniques and	No single best-performing method identified

Title	Author(s)	Methods / Algorithms	Dataset	Key Findings	Limitations
Techniques in LLMs		mitigation techniques		categorizes them systematically	
LLM-Check: Investigating Detection of Hallucinations	G. Sriramanan et al.	Hidden state, attention maps, and logit-based detection	Multiple datasets	Efficient detection with up to 45x speed improvement over baselines	Requires access to internal model architecture
Hallucination Detection and Mitigation: An Investigation	J. Luo et al.	Comprehensive review of detection and mitigation approaches	Various benchmarks	Hybrid approaches combining detection + mitigation outperform standalone methods	Lacks domain-specific evaluation
A Systematic Literature Review of Hallucinations in LLMs	C. Woesle et al.	PRISMA-based systematic review and classification framework	125 research papers	Only ~8.8% works implement integrated solutions; highlights research gaps	Lack of unified evaluation standards
Why Language Models Hallucinate	A. T. Kalai et al.	Theoretical analysis using learning theory	Conceptual	Hallucinations arise due to training objectives rewarding guessing over uncertainty	No direct practical mitigation solution
Beyond Binary Rewards: Training LMs for Uncertainty	M. Damani et al.	RLCR (Reinforcement Learning with Calibration Rewards)	QA datasets	Improves confidence calibration and reduces overconfident hallucinations	Requires retraining and additional computation
Uncertainty-Aware Fusion (UAF)	P. Dey et al.	Ensemble learning with uncertainty-based fusion	QA datasets	Improves factual accuracy (~8%) by combining multiple models	High computational cost and latency
Orchestrating Consensus Strategies	J. Nambiar et al.	Multi-model consensus and agreement-based selection	Chatbot outputs	Consensus among models reduces hallucination and improves reliability	Slow and resource-intensive
Mitigating Hallucinations	H.-T. Ho et al.	RAG with knowledge graph	Educational datasets	Improves domain-specific	Limited generalization

Title	Author(s)	Methods / Algorithms	Dataset	Key Findings	Limitations
in Educational LLMs		integration		factual accuracy in educational.	outside domain
Detecting & Reducing Hallucinations with Metamorphic Testing (DrHall)	W. Wu et al.	Metamorphic testing, consistency checking, multi-path voting	FactHalluQA, Refactory, LeetCode	Achieves ~0.85 F1 detection and improves correction rate using consistency	Requires multiple test cases and queries
MH-PEFT: Mitigating Hallucinations in LVLMs	F. Li	Data augmentation + parameter-efficient fine-tuning (PEFT)	Multimodal datasets	Improves multimodal alignment and reduces hallucination with lightweight training	Requires training and multimodal data
CPR: Curative Prompt Refinement	J.-W. Shim et al.	Prompt cleaning, paraphrasing, and description generation using SLM	Wikipedia, MQR, Wikidata	Achieves ~90–96% improvement in output quality by refining prompts	Depends on input preprocessing quality

2.1 Overview of Existing Solutions

Lots of research has gone into the problem of Large Language Models (LLMs) “hallucinating” (making things up), and a lot of ways to find and lessen this tendency have been suggested. One common way to spot a hallucination is to look at semantic consistency; you give the model the same question asked in different words and if the answers are very different, it probably fabricated something. Or, a method called self-consistency has the model answer the same question many times and then looks at the answers - because invented information won’t usually be the same each time.

Another important trick for finding hallucinations is self-contradiction analysis, which means looking for statements inside one answer that say opposite things. Because both of those contradictory statements can’t be true, this is a good way of showing up information the model has made up. While these ways of finding issues do tell you which outputs to not trust, they might require more computing power or you might need to ask the model the same thing more than once.

To fix the problem, Retrieval-Augmented Generation (RAG) is an approach that gets information from outside sources to be more factually correct. Also, carefully written prompts and including important background in the question (context injection) can lead the model to be more accurate by being very specific with what is asked. Ensemble methods (putting together answers from several models) and figuring out how sure the model is of its answer (uncertainty estimation) to lessen very confident, but wrong, answers are also used. In general, research shows you get a better outcome if you use multiple solutions together, rather than just one.

2.2 Identified Challenges

The problem of hallucinations in LLMs is still present despite great advancements in recent years. It is said that distinguishing between a fabrication and a factual statement is particularly difficult. Because a hallucination may seem coherent and relevant regarding the context of the discussion, it may not seem like a fabrication.

Another problem is that there is no verification mechanism for LLMs like the one that exists for Factuality. Because of this, LLMs will generate a response that would seem plausible when it may not even contain true information.

Most of the advanced fabrication detection methods require multiple queries to the model and/or require a data set and this suggests higher costs and less efficiency, particularly in real-time systems.

Domain specific, especially in the area of health care, continues to be a problem. General purpose methods that may not be suitable for domains that require meticulous accuracy.

Another problem is the siloed nature of the hallucinatory detection evaluation frameworks. With no standard evaluation framework, it is difficult to assess and compare the available methods to put forth the best of the techniques.

2.3 Opportunities for Improvement

Regarding existing methodologies' shortcomings, there are areas for potential improvement and innovation.

One such potential improvement concerns developing measures that detect hallucinations without requiring multiple outputs or extensive computation that can be performed in a single pass. One such measure that has the potential to be used in a single pass computation is based on internal model signals (e.g. attention, confidence score) within the model.

Additionally, there is a need for the development of tests for hallucinations in defined areas, especially in the field of healthcare. This can be achieved through the incorporation of medical verified datasets and that healthcare personal knowledge in order to improve accuracy and reliability.

Moreover, the development of detection and mitigation hybrid systems is relatively unexplored Still, the development of systems in which a hallucination is first detected by consistency checks and then corrected through retrieval-based correction is likely to be positively received.

Additionally, the introduction of uncertainty and confidence models are expected to improve the perceived and actual trust of the systems as well as enhance their transparency.

Finally, there needs to be established evaluation methods and datasets that can be used to measure the effectiveness of various techniques to streamline and guide future development efforts.

2.4 Summary of Findings

The literature review indicates that hallucination is a fundamental and persistent issue in LLMs, arising from their probabilistic nature and training processes. While several detection and mitigation techniques have been proposed, each method has its own limitations.

Detection methods such as semantic consistency, self-consistency, and contradiction analysis are useful but may require additional computation. Mitigation approaches like retrieval augmentation, prompt engineering, and ensemble methods improve accuracy but are not fully reliable on their own.

Overall, the most effective strategy is to use a combination of detection and mitigation techniques, tailored to specific application domains. This highlights the need for integrated systems that can both identify and reduce hallucinations, especially in critical areas like healthcare.

3. Problem Definition

3.1 Project Scope

The scope of this project focuses on the detection and mitigation of hallucinations in Large Language Models (LLMs), with a specific emphasis on healthcare-related applications where accuracy is critical. The system aims to analyze responses generated by LLMs and determine whether the information is factually correct, inconsistent, or potentially hallucinated.

The project involves implementing multiple detection techniques such as semantic consistency checking, contradiction analysis, and confidence estimation, which help identify unreliable outputs. In addition, mitigation strategies such as retrieval-based verification, prompt refinement, and ensemble approaches are incorporated to reduce the occurrence of hallucinations and improve response quality.

A prototype system is developed that integrates both detection and mitigation modules into a single pipeline. The system processes user queries, generates responses using an LLM, evaluates the outputs for hallucinations, and applies corrective measures where necessary.

However, the scope is limited to a prototype-level implementation and does not include training large-scale models from scratch due to computational constraints. The project primarily focuses on text-based LLMs and selected datasets, particularly in the healthcare domain. The system may not fully generalize to all real-world scenarios or multimodal applications.

3.2 Objectives

The main objectives of this project are as follows:

- To study and analyze the causes, types, and impact of hallucinations in Large Language Models
- To design and implement effective hallucination detection techniques using consistency, contradiction, and confidence-based methods
- To develop and integrate mitigation strategies such as retrieval-based grounding and prompt optimization
- To build a prototype system that combines detection and mitigation in a unified framework
- To evaluate the system's performance using standard metrics such as accuracy, precision, recall, and F1-score
- To improve the reliability, interpretability, and trustworthiness of LLM-generated responses

The overall objective is to create a system that not only identifies hallucinations but also actively reduces their occurrence, making LLMs safer for practical use.

3.3 Deliverables

The expected deliverables of this project include:

- A functional prototype system capable of detecting and mitigating hallucinations in LLM outputs
- A structured dataset collection and preprocessing pipeline, including healthcare-related queries and test cases
- Implementation of detection algorithms (consistency checks, contradiction detection, confidence scoring)
- Implementation of mitigation techniques (retrieval augmentation, prompt refinement, ensemble methods)
- A detailed performance evaluation report, including metrics such as accuracy, precision, recall, and F1-score
- A comprehensive project report documenting the methodology, system design, results, and conclusions

These deliverables collectively demonstrate the feasibility and effectiveness of the proposed solution.

3.4 Problem Statement

Large Language Models (LLMs) are increasingly used in real-world applications due to their ability to generate fluent and contextually relevant text. However, they often produce hallucinated outputs, which are factually incorrect, fabricated, or misleading despite appearing convincing. This issue arises because LLMs rely on probabilistic text generation rather than factual verification.

The presence of hallucinations poses significant challenges, particularly in high-stakes domains such as healthcare, where incorrect information can lead to serious consequences, including misdiagnosis or improper treatment recommendations. Furthermore, hallucinations reduce user trust and limit the practical adoption of LLM-based systems.

Therefore, the problem addressed in this project can be stated as:

“How can hallucinations in Large Language Model outputs be accurately detected and effectively mitigated using scalable and reliable techniques, in order to improve the overall reliability and trustworthiness of generated responses, especially in healthcare applications?”

4. Proposal Development

4.1 Research Findings

Based on the literature review, hallucinations in Large Language Models (LLMs) are identified as a systemic issue caused by their probabilistic text generation nature. LLMs are designed to predict the most likely sequence of words rather than verify factual correctness, which leads to the generation of plausible but incorrect information.

Research highlights several key causes of hallucinations, including lack of grounding in real-world data, inconsistencies in training datasets, and model overconfidence when handling uncertain inputs. Studies also show that hallucinations can occur in various forms such as factual errors, fabricated details, and internal contradictions.

Existing research suggests that hallucination detection can be achieved using techniques such as semantic consistency checks, contradiction detection, and confidence estimation, while mitigation can be performed using retrieval-based verification, prompt engineering, and ensemble methods. However, no single approach is sufficient to completely eliminate hallucinations.

A major insight from recent studies is that hybrid approaches combining detection and mitigation techniques provide better performance and reliability. This forms the basis of the proposed system in this project.

4.2 Project Objectives

Based on these findings, the proposed system aims to achieve the following objectives:

- To design a system capable of detecting hallucinations in LLM-generated responses
- To implement effective mitigation techniques that reduce incorrect outputs
- To integrate detection and mitigation into a unified and efficient framework
- To improve the accuracy, reliability, and trustworthiness of generated responses
- To apply the system in healthcare-related scenarios, where correctness is critical

The overall goal is not only to identify hallucinations but also to actively reduce their occurrence and improve output quality.

4.3 Methodology

The methodology follows a hybrid pipeline approach, where both detection and mitigation techniques are integrated into a single system. The process consists of the following stages:

1. Input Processing
   The system accepts user queries, particularly healthcare-related questions, and prepares them for processing.
2. Response Generation
   The input query is passed to an LLM to generate an initial response.
3. Hallucination Detection
   The generated response is analyzed to identify potential hallucinations.
4. Hallucination Mitigation
   If hallucination is detected, corrective techniques are applied to improve the response.
5. Final Output Generation
   The corrected or validated response is presented to the user with improved reliability.

This structured pipeline ensures that hallucinations are both detected and minimized effectively.

4.3.1 Data Collection Methods

The data used in this project is collected from multiple sources to ensure diversity and reliability:

- Public Datasets:
  Standard datasets such as healthcare QA datasets and benchmark datasets are used for evaluation.
- Synthetic Data Generation:
  LLMs are used to generate both correct and hallucinated responses to create test cases.
- Research-Based Datasets:
  Datasets referenced in research papers (e.g., TruthfulQA, fact-based QA datasets) are used for validation.
- Manual Test Cases:
  Custom healthcare-related queries are designed to simulate real-world scenarios.

The collected data is preprocessed and structured to support both detection and mitigation experiments.

4.3.2 Analytical Techniques

The system uses multiple analytical techniques to evaluate and improve LLM outputs:

- Semantic Consistency Analysis
  Compares outputs for similar or paraphrased inputs. Large variations indicate potential hallucination.
- Contradiction Detection
  Identifies conflicting or logically inconsistent statements within the response.

- Confidence Estimation
  Analyzes model confidence scores to detect uncertain or unreliable outputs.
- Retrieval-Based Verification
  Cross-checks responses with external knowledge sources to ensure factual correctness.
- Prompt Refinement Analysis
  Improves input queries to reduce ambiguity and enhance output accuracy.
- Ensemble Validation Techniques
  Uses multiple model outputs and selects the most consistent response.

These techniques collectively improve both detection accuracy and mitigation effectiveness.

4.4 Expected Outcomes

The proposed system is expected to achieve the following outcomes:

- Improved factual accuracy of LLM-generated responses
- Effective detection of hallucinated or unreliable outputs
- Reduction in incorrect or misleading information through mitigation strategies
- Enhanced trustworthiness and reliability of LLM-based systems
- Better performance in handling healthcare-related queries

The results are expected to demonstrate that combining detection and mitigation techniques significantly improves system performance.

4.5 Timeline and Milestones

The project is planned over a period of approximately 6 weeks, with the following milestones:

Phase	Duration	Description
Literature Review	2 weeks	Study research papers and existing methods
Data Collection	1 week	Gather medical datasets and prepare test cases
System Design	1 week	Design system architecture and methodology
Prototype Development	2 weeks	Develop hallucination detection modules
Integration	1 week	Integrate LLM, dataset, and detection system
Testing & Evaluation	1 week	Evaluate performance using defined metrics
Documentation	1 week	Prepare final report, results, and presentation

5. Prototype Design

5.1 Design Specifications

The proposed system is designed to detect hallucinations in Large Language Model (LLM) responses for healthcare-related queries. The system follows a modular architecture consisting of the following components:

1. Data Module
    - Uses medical datasets such as PubMedQA and HealthCareMagic
    - Builds a combined knowledge base for verification
2. Retrieval Module
    - Retrieves relevant context using semantic similarity
    - Uses SentenceTransformer embeddings
3. Generation Module
    - Uses two models:
        - Student Model: *flan-t5-small*
        - Teacher Model: *flan-t5-base*
4. Hallucination Detection Module
    - Compares outputs using cosine similarity
    - Detects inconsistencies with knowledge base
5. Mitigation Module
    - Replaces or refines hallucinated responses

The system is implemented in Python and integrates NLP models with similarity-based verification.

5.2 Functionality Requirements

The system performs the following key functions:

- Accepts healthcare-related user queries
- Retrieves relevant medical knowledge from datasets
- Generates responses using LLMs
- Computes similarity scores between:
    - student vs teacher response
    - student vs context
- Detects hallucinated responses
- Applies mitigation strategies to improve output quality
- Provides final verified response

5.3 Feasibility & Scalability Analysis

Feasibility

- Uses open-source models → low cost
- Runs on CPU/GPU/Mac MPS → accessible
- Uses existing datasets → no need for manual data collection

Scalability

- Can be extended to:
    - larger datasets
    - real hospital databases
    - real-time API systems

The modular design allows easy upgrades such as:

- better LLMs
- improved retrieval systems

5.4 Design Validation Criteria

The system is validated based on:

- Accuracy of generated responses
- Similarity with ground truth answers
- Ability to detect hallucinated outputs
- Improvement after mitigation

6. Prototyping

6.1 Proof-of-Concept Development

A proof-of-concept (PoC) system was developed to demonstrate the detection and mitigation of hallucinations in Large Language Models (LLMs) for healthcare-related queries. The prototype integrates multiple stages, including dataset preparation, semantic retrieval, response generation, hallucination detection, and output refinement.

The system uses medical datasets such as PubMedQA and HealthCareMagic to construct a combined knowledge base containing verified medical information. This knowledge base includes structured entries consisting of questions, contextual information, and corresponding answers. It acts as a reliable reference for validating responses generated by the language models.

A dual-model architecture is implemented using a student model (flan-t5-small) and a teacher model (flan-t5-base)**.** The student model generates initial responses quickly, while the teacher model produces more accurate and reliable outputs. This combination allows the system to compare responses and identify inconsistencies.

Additionally, a semantic retrieval mechanism is used to extract relevant context from the knowledge base using sentence embeddings. This ensures that the generated responses are grounded in domain-specific medical knowledge. The PoC successfully demonstrates a complete pipeline for identifying hallucinated outputs and improving response reliability.

6.2 Key Functionalities Implementation

The prototype system implements several key functionalities to enable hallucination detection and mitigation.

The first component is knowledge base construction, where data from PubMedQA and HealthCareMagic datasets are combined into a unified structure. Each entry contains a question, context, answer, and source label, allowing the system to handle both research-based and patient-oriented queries.

The second component is semantic retrieval, which uses embedding-based similarity to identify the most relevant entries from the knowledge base. The top-k relevant contexts are selected and used to guide response generation.

The third component is response generation, where both student and teacher models generate answers based on the query and retrieved context. Domain-specific prompts are used to improve the quality and relevance of responses.

The fourth component is hallucination detection, which is based on similarity comparisons. The system evaluates:

- similarity between student and teacher responses
- similarity between student response and retrieved context

If these similarity scores are below predefined thresholds, or if the response is insufficient, it is classified as hallucinated.

The final component is mitigation, where a three-level strategy is applied:

- replacing the response with the teacher output
- blending student and teacher responses
- accepting the student response if confidence is high

This approach ensures improved accuracy and reliability of the final output.

6.3 Tools & Technologies Used

The prototype is implemented using Python along with several libraries and frameworks. Hugging Face Transformers are used for loading and running language models, while SentenceTransformers are used for generating embeddings and computing similarity scores. PyTorch is used for model execution and supports CPU, GPU, and Apple MPS acceleration.

The datasets library is used to load PubMedQA and HealthCareMagic datasets, and Matplotlib is used for visualization of results and performance analysis.

6.4 System Workflow

The workflow of the prototype system is as follows:

1. User inputs a healthcare-related query
2. Relevant context is retrieved from the knowledge base
3. Student and teacher models generate responses
4. Similarity scores are computed
5. Hallucination is detected based on thresholds
6. Mitigation strategy is applied
7. Final verified response is returned

6.5 Advantages and Limitations

The proposed prototype improves the reliability of LLM outputs by grounding responses in verified medical data and reducing hallucinations using similarity-based detection. It is cost-effective and can be extended to larger datasets or real-world healthcare systems.

However, the system depends on the quality of the dataset and requires proper tuning of similarity thresholds. It is also currently limited to text-based queries and may need further optimization for real-time deployment.

7. Testing and Validation

7.1 Test Plan & Procedures

The proposed system was tested using healthcare-related queries from two datasets: PubMedQA and HealthCareMagic. These datasets provide both research-oriented and patient-oriented medical questions, allowing comprehensive evaluation of the system.

The testing procedure followed a structured approach:

1. A query was selected from the test dataset
2. Relevant context was retrieved from the knowledge base using semantic similarity
3. Responses were generated using both the student and teacher models
4. Similarity scores were calculated between:
    - student and teacher responses
    - student response and retrieved context
5. The system evaluated whether the response was hallucinated based on predefined thresholds
6. A mitigation strategy was applied if hallucination was detected
7. The final response was compared with the ground truth answer

This process was repeated for multiple test cases to ensure consistent evaluation across different types of medical queries.

7.2 Performance Metrics

The system was evaluated using the following performance metrics:

- Cosine Similarity**:**
  Measures semantic similarity between generated responses and ground truth answers
- Accuracy:
  Percentage of responses that are considered correct based on similarity thresholds
- Hallucination Rate:
  Percentage of responses identified as hallucinated
- Detection Rate:
  Number of hallucinated responses correctly detected by the system

These metrics help assess both the quality of generated responses and the effectiveness of hallucination detection.

7.3 Experimental Results

The evaluation results show that the system is effective in detecting and mitigating hallucinations. The student model, when used alone, often produces responses with lower similarity to the ground truth. However, the teacher model improves response quality due to its higher capacity.

After applying the mitigation strategy, the final responses show improved similarity scores and reduced hallucination rates. The system successfully identifies low-confidence or incorrect outputs and replaces or refines them using the teacher model.

The results also indicate that the retrieval of relevant context plays a crucial role in improving response accuracy. Queries with better contextual matches show higher similarity scores and fewer hallucinations.

7.4 Iteration Results

Several improvements were made during system development through iterative testing:

- Increasing the number of retrieved contexts (top-k) improved grounding
- Using domain-specific prompts enhanced response quality
- Adjusting similarity thresholds improved detection accuracy
- Implementing multi-level mitigation improved final output reliability

These iterations resulted in a noticeable improvement in system performance, including higher accuracy and lower hallucination rates.

7.5 Summary of Validation

Overall, the system demonstrates effective performance in detecting hallucinations in healthcare-related queries. The combination of semantic retrieval, dual-model comparison, and mitigation strategies improves both accuracy and reliability.

The testing results validate that the proposed approach can be used as a foundation for building trustworthy AI systems in healthcare applications.

8. Data Collection & Analysis

8.1 Testing Data Methodology

The system was evaluated using healthcare-related datasets, namely PubMedQA and HealthCareMagic. These datasets were selected to cover both research-based and patient-oriented medical queries.

A subset of data was used for testing, where each sample consisted of a question and its corresponding ground truth answer. The evaluation process involved generating responses using the student and teacher models and comparing them with the ground truth.

For each query:

- Relevant context was retrieved from the knowledge base
- Responses were generated by both models
- Similarity scores were computed
- Hallucination detection and mitigation were applied

This structured methodology ensured consistent and fair evaluation across all test cases.

8.2 Key Findings & Metrics

The analysis of the system revealed several important findings:

- The student model alone often produced responses with lower accuracy and higher hallucination rates
- The teacher model generated more accurate and reliable responses due to its larger architecture
- The final mitigated output showed improved performance compared to both models

Key metrics observed:

- Accuracy increased after applying mitigation
- Hallucination rate decreased significantly
- Similarity scores improved, indicating better alignment with ground truth

It was also observed that:

- Queries with better retrieved context had higher accuracy
- Domain-specific prompting improved response quality
- Multi-level mitigation helped reduce incorrect outputs

8.3 Data Visualization

The system generates graphical representations of performance using Matplotlib. These visualizations provide a clear understanding of the system's effectiveness.

The main graphs include:

- Accuracy vs Hallucination Rate:
  Shows the trade-off between correct responses and hallucinated outputs
- Mean Similarity Comparison:
  Compares student, teacher, and final responses
- Per-Question Analysis:
  Displays similarity scores for each test query

From the graphs, it can be observed that:

- The final output consistently performs better than the student model
- Hallucination detection successfully identifies low-confidence responses
- Mitigation improves overall system reliability

These visualizations support the conclusion that the proposed system enhances the accuracy and trustworthiness of LLM outputs.

9. Patent Search

9.1 Relevant Patents Review

A patent search was conducted to analyze existing intellectual property related to hallucination detection, fact verification, and reliability improvement in Large Language Models (LLMs) and AI systems.

Several patents and technologies focus on improving the trustworthiness of AI-generated content, particularly in domains such as healthcare and decision support systems.

1. AI-based Fact Verification Systems

Many patents describe systems that verify generated outputs using external knowledge bases or structured data sources. These systems compare model-generated responses with trusted databases to ensure factual correctness.

2. Ensemble-based AI Verification

Some patents propose the use of multiple models to generate and validate responses. These systems rely on consensus or voting mechanisms to reduce incorrect outputs, similar to multi-model verification approaches.

3. Confidence-based Filtering Systems

Certain patented approaches use confidence scores or probability measures to determine whether a response is reliable. Low-confidence outputs are flagged or corrected before being presented to the user.

4. Domain-Specific AI Systems (Healthcare)

In healthcare applications, patents emphasize the use of verified medical databases and clinical guidelines to ensure safe and accurate responses. These systems aim to reduce risks associated with incorrect medical advice.

9.2 IP Gap Analysis

Based on the patent review, several gaps and opportunities for improvement have been identified:

- Most existing systems rely on complex architectures or multiple models, making them computationally expensive
- Limited focus on lightweight, real-time hallucination detection systems
- Few systems combine semantic similarity-based verification with dual-model comparison
- Existing healthcare AI systems often lack dynamic hallucination detection mechanisms

The proposed system addresses these gaps by:

- Using a cost-effective dual-model approach (student + teacher)
- Applying semantic similarity-based detection instead of heavy ensemble methods
- Integrating context retrieval from domain-specific datasets
- Providing a multi-level mitigation strategy to improve response reliability

Conclusion of Patent Analysis

The patent study indicates that while significant research has been conducted in AI reliability and verification, there is still scope for developing efficient and scalable hallucination detection systems. The proposed project contributes to this area by offering a simple, effective, and domain-specific solution suitable for healthcare applications.

10. Conclusion

10.1 Summary

This project focused on detecting hallucinations in Large Language Models (LLMs) for healthcare-related queries. A prototype system was developed that combines semantic retrieval, dual-model response generation, similarity-based hallucination detection, and multi-level mitigation strategies.

The results show that LLMs can generate plausible but incorrect responses, especially when not grounded in reliable knowledge sources. The proposed system successfully identifies such hallucinations by comparing model outputs with contextual information and reference responses. The use of a teacher model and knowledge-based retrieval significantly improves the accuracy and reliability of generated answers.

Overall, the system demonstrates that hallucination detection can enhance the trustworthiness of AI systems, particularly in sensitive domains such as healthcare where accuracy is critical.

10.2 Next Steps

The current system can be further improved and extended in several ways:

- Integration with real-time medical databases and APIs for updated knowledge
- Use of advanced and domain-specific LLMs for higher accuracy
- Expansion to handle multi-modal inputs such as medical images and reports
- Optimization for real-time deployment in hospital environments
- Incorporation of feedback mechanisms for continuous learning and improvement

Future work can focus on developing a fully scalable and production-ready system that ensures safe, reliable, and efficient AI-assisted healthcare solutions.

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